Electronic payment systems

P A Putland, C Ward, A Jackson and C Trollope

There has been slow but steady progress in the technology behind electronic payment systems in the last two years, since most effort has concentrated on integration between the various components needed in the complete purchasing process to produce end-to-end solutions rather than in the actual payment systems themselves. This paper will give a brief review of the technological progress, mentioning the issues that have been addressed during the last two years, and provide a brief overview of the types of payment schemes currently being used in the market-place.

1. Introduction

We are still awaiting the expected explosion in commerce over the Internet in Europe, although recent reports suggest that on-line commerce is starting to accelerate in the US. There are some well-known companies, such as Dell, Cisco and Amazon, which generate huge revenues from their Web sites, but recent figures show that Christmas 1998 was perhaps the start of the movement of on-line sales from a few well-known sites into a more mass-market phenomenon (traffic to AOL's shopping channel increased by 350%, and Boston Consulting Group say on-line sales tripled to $13 billion [1]). Indeed, of those who spent money on-line, Zona research reported that over 60% of buyers expect to spend more money on-line in the coming year.

Despite these promising statistics, on-line payment of items is still not as widespread as predicted, with Infobead [2] producing a recent report quoting 'non-trust of the payment method' as one of the top reasons why users do not buy on-line. Also, some of the earlier innovative technologies have not had the take-up expected. Digicash has gone bankrupt, digital certificates have not quite taken off for consumer use, and Mondex is making slow but steady progress. While the penetration of PCs is increasing, the penetration of other enabling technologies — such as card readers — into the general population has been slow. However, we are at the stage where many of the ideas and concepts described a few years ago are now in trial or service. Payment options available for consumers on the Internet range from sub-currency (i.e. less than one penny) through to thousands of pounds. The range of technologies to support these include client wallets holding tokens or linked to accounts, server-held accounts with no client software needed, and smart-card-based payment systems.

A full review of electronic payment systems has been published previously [3], but for completeness a summary is included here. The various types of payment transaction can be broadly broken down into card, cheque or cash. Card transactions can occur very easily over the Internet by simply asking the customer to enter the card details on a Web page, and confidentiality is maintained by using secure sockets layer (SSL). This method does not provide a full audit trail (needed in case of dispute resolution) which led to the introduction of commerce platforms that not only captured the payment information, but also had proof of audit and delivery components. Mastercard and Visa have jointly proposed a full cryptographic solution called secure electronic transaction (SET) [4], which provides full audit trails, while ensuring that the card details are not disclosed to retailers. Electronic cheque transactions are beginning to occur in the USA, but there are no widely available systems in the UK at present. Digital cash technologies have also not taken off as anticipated by the suppliers, with most effort focusing on micropayments (systems that aggregate small payments together to make the service economically viable). The majority of systems used on the Internet at present to facilitate low-value payments are simple account-based systems, with purchases being charged to a pre- or post-paid account.

Since that paper [3] was published, there have been few technological changes, but progress has been made in consolidation and integration within the industry. This paper will briefly look at the current status of electronic payment systems, and progress in the last two years to widen the appeal and use of such systems.

2. The need for electronic payments

The term electronic payment covers what is in effect a simpler model of the more complex mainstream enterprise model of 'billing'. In newer telecommunications services, pre-payment is becoming more important. Not only as phone cards for public telephones, but also for mobile phones where they offer the customer a degree of control over expenditure, and in general micropayment systems where pre-payment considerably reduces the risk model. From the billing perspective, pre-payment removes most of what would be seen as traditional billing functionality, thereby reducing costs and increasing speed to market.

Also, there is no common approach to charging for Internet and eCommerce services. Generally, most Internet service providers (ISPs) have gone for the simplicity of a subscription-based approach. However, in the UK and most European countries, normal PSTN or ISDN access charges are also payable and these are usage based. In the US the situation is reversed with most access being free and subscriptions being the norm. One issue is that the Internet protocol does not support usage monitoring, although work is under way to find a means to overcome this. What is probable is that the range of payment methods needed to support the various business models for new telecommunications services is likely to increase rather than decrease.

3. The purchasing process

In order for electronic payments to work it is necessary that many different things occur, although the consumer and merchant may only be presented with a simple electronic payment model. The requirements, and hence technical solutions, to on-line payments differ depending on the circumstances, the amount being paid, and how the payment fits into the overall purchasing process. It is worth making the point at this stage that although this paper is about payment mechanisms, most of the mechanisms discussed so far actually cover a broader role. All micropayment services at the moment provide for the payment of soft goods that are delivered immediately to the end user. In order for them to provide the aggregation, they all have customer registration processes, as well as the ability to deliver items securely to the user. With this in mind, the only true payment protocols are Mondex, which is a true digital cash, and the W3C MPTP protocol (see section 6).

The complete purchasing process is beyond the scope of this paper, and protocols such as OTP (open trading protocol [5]) aim to separate out the various parts of a transaction, leaving a pure payment mechanism. However, at present, most so-called payment mechanisms actually include other components, and the range that could be relevant for a payment system include the following.

- Order entry

  Using a simple Web form or catalogue system, the goods are presented and selected. The channel for this is usually direct to the consumer, although other parties, such as call centres, may be involved to allow a customer service representative to enter the details.

- Pricing

  In the simple case, there is a direct relationship between goods/services and prices, though tariffs could vary according to supply and demand. Usage-based charging and recurring charges for service are struggling to assert themselves in many areas.

- Taxation and discounting

  Currently simple models on a per-item basis are used aimed at the consumer and small business market. Traditionally it is the larger corporations that require more complexity in this area.

- Invoicing

  Electronic payments are usually associated with electronic invoices, as e-mail, or through systems that provide an on-line view.

- Collections

  A collection is assumed completed once authorisation from the financial service provider for the charge is confirmed. This is a problem when there are recurring charges with credit cards (recurring charges are not allowed under UK SWITCH rules) with the expiry date or even card number changing over time. This and ‘cardholder not present’ rules, resulting in possible chargebacks, create a requirement for debt-management facilities, to cater for those occasions when payment is not collected.

4. Consumer transactions

These can very broadly be broken down into two types — those for low value that users will treat in the same way as physical cash, and those with the same attributes as larger value purchases.

4.1 Low-value transactions

Low-value transactions are characterised by the need for protocols that do not require vast computing or storage resources, and in some cases do not even need to offer an audit trail (if the cost of the transaction is 0.01p, the cost of providing the audit trail, with associated customer support systems, could cost more than the value of the transaction). All low-value systems work by having one large settlement act as payment for a number of smaller purchases. There are several players in this market-place offering payments as low as one penny, and BT has tested a trial system during the last year.

One set of players (such as Cybercash, KLELine, Cybank) fall into the software wallet camp where the user downloads a software wallet on to their PC, and the wallet holds the user’s account or value details. The wallets use encryption to secure the information for storage and repeated transmission over the network.

The other set of players (such as OpenMarket, BTArray, Worldpay) use secure servers to hold the user details, and rely on SSL encryption to provide the security during the initial registration, and then the personal information is not sent over the network during each transaction. This approach has the advantage that users can then access the system from any client, rather than be restricted to a PC containing the software wallet.

With the likely introduction of smart-card technology over the coming years, it is likely that the need for software wallets will decrease, and the technical solutions offered into the market-place may start converging (Cybercash are already starting to move away from client-wallet technology towards strong authentication based upon a smart-card stored ‘secret’).

4.2 High-value transactions

High-value transactions are typically carried out by consumers using a credit or debit card, and the technologies that support these have been covered above. Since then the main progress in credit card payments on the Internet has been in three areas.

- Payment APIs

  The integration of the payment APIs from the main players into the products provided by the on-line catalogue suppliers — this means that if a retailer buys a solution to build an on-line catalogue from providers such as Intershop or iCAT, etc, there is a choice of payment API already provided to ease integration, making it very simple for retailers to accept credit card orders.

- SET protocol

  The strong foothold gained by using simple SSL, or client wallet, technology to encrypt the credit card details, with various back-end solutions to provide the rest of the purchase process — two years ago it was thought that the SET [3] protocol would be deployed and widely used by now. Delays in the specification, the cost of implementation and the lack of a public-key infrastructure (PKI) to support the necessary certificate infrastructure have all led to the low use of SET. It is now unlikely that SET will be used widely before smart cards become prevalent.

- User wallets

  The incorporation of a wallet into the standard desktop components — unsurprisingly, Microsoft is the major player in this area, and their current browser (Internet Explorer 4) has a wallet with credit card payment functionality built in. The user experience of buying an item using an IE4 browser from a retailer using the Microsoft Commerce Site Server is very good, with all interactions being carried out within the browser. Further, the Microsoft wallet has a COM architecture that allows additions of other payment modules. By default, the Microsoft Wallet includes a Credit Card Payment Module, but other payment application vendors can extend the wallet to support additional payment methods by developing a payment module. For example, CyberCash have developed a module that supports their Cybercoin micropayments solution, enabling the Microsoft wallet user or merchant to add Cybercoin functionality to the user wallet.

5. Business-to-business transactions

At this point it is worth mentioning business-to-business transactions. These have different characteristics to consumer transactions, both in the general purchasing model (businesses prefer to be invoiced then pay later) and the authority needed (in large companies, there tend to be internal processes to go through before a payment can be made). In these respects, most value can be added by providing the integration between the payment method and the accounting and invoicing packages used with organisations, rather than just the automation of the actual payment method itself.

6. Industry consortia

There are several industry consortia producing various specifications for generic trading, or specific payment protocols. This is needed, as the current profligacy of proprietary solutions is fragmenting the market, and in some cases preventing retailers going on-line or users making purchases as they neither know which technology to use, nor want to have to use several. As well as the SET protocol mentioned above, at present there appear to be three other key consortia.

- Internet Engineering Task Force (IETF)

  The IETF has now taken over the open trading protocol (OTP) from the original OTP Consortium, in which Mondex was a key player. As Mondex is a true payment-only protocol, there needs to be higher level trading protocols that provide the environment in which the payment can take place. These higher level protocols cover issues such as negotiation of terms, invoicing, receipting, delivery, etc. As some of the main providers of electronic payment systems (Cybercash, Mondex, Globe ID) were involved in the original OTP specifications, it is likely that the non-payment elements of their systems will migrate to a common set of protocols.

- W3C

  In November 1995 the World Wide Web Consortium (W3C) produced a working draft (version 0.1) for a micropayment transfer protocol (MPTP). There has been recent activity in the W3C micropayment group to define a payment API. This work is predicated on the existence of a wallet on the user's client, but, with the expected use of XML, this could have an impact in that it may define a standard payments tag that can be embedded in the HTML pages that a retailer publishes. This tag would then be interpreted by the client to initiate a payment using an appropriate payment mechanism.

- Open Financial Exchange (OFX)

  The OFX standard grew out of a joint initiative by CheckFree, Intuit and Microsoft in 1996. OFX bill presentation is a part of the standard, which allows corporates to develop a system that can interface with partners and bill consolidators. CheckFree has set up an E-Bill partner programme to encourage wider adoption of Internet billing. It currently offers e-billing and payment services to over 2m customers in the USA and electronic links to 1000 merchants. AT&T joined with CheckFree in mid-1998 to offer residential customers the ability to view and pay their bills over the Internet using CheckFree's electronic systems.

7. Smart cards

Smart cards are covered in detail elsewhere [6], but for completeness their role in payments is briefly mentioned here. Although there are many electronic purse schemes running in various parts of the world [6], there are no national e-purse schemes currently operating in the UK, although both Mondex and Visa are running trial systems in the UK, with BT providing the payphones for both.

7.1 Mondex

Mondex e-cash is stored in the Mondex e-purse providing an alternative to notes and coins. Mondex e-cash is anonymous and relies on smart-card security to prevent electronic fraud. The Mondex e-purse allows card-to-card value transfers in a variety of ways, e.g. an electronic handheld wallet, BT Mondex telephones and automatic teller machines (ATMs). BT played a major role in the Mondex pilot scheme in Swindon to develop a street payphone and residential telephone to accept the Mondex e-purse for payment and value transfer. As a result in-depth and practical knowledge of the Mondex e-purse was acquired which was used to develop an electronic payment over the Internet demonstrator.

Work was carried out at BT Laboratories to produce a demonstrator, which transferred Mondex value between two PCs using TCP/IP protocols over the Internet. The demonstrator was integrated with BT's experimental trading platform [7], so that electronic goods could be purchased using Mondex e-cash. The demonstrator adopts the latest Mondex card technology, which will be used for all new Mondex cards issued world-wide. The BT Internet demonstrator performs a value transfer in less than 9 seconds, which is better than the estimates from within the industry for a Mondex value transfer using Internet technology.

7.2 Visa Cash

Visa Cash in the UK is on trial in Leeds where BT is providing payphones, which accept Visa Cash e-cash. Visa Cash to the cardholder may behave like Mondex e-cash, but it is actually more akin to a credit or debit card purchase where records are generated for each transaction, and are passed back to the bank issuing the currency. This means that Visa Cash is traceable whereas Mondex is untraceable. The Visa Cash e-purse can be added to a debit or credit card or issued as a separate card. Visa Cash is thus a competitor to Mondex.

8. Currency issues

The use of multi-currencies is perhaps the area where European payment technology leads the world. In view of the dominance internationally of the dollar, most USA-based payment system suppliers have created single currency solutions based on the dollar. Because of the various different currencies within Europe, several European systems, such as KLELine and Worldpay, provide multi-currency support, with real-time exchange rate information available. The Euro has recently entered life as an electronic currency within Europe, and is already supported by suppliers of multi-currency systems. As the Euro is only usable in electronic transactions, since no notes or coins are yet available, it is possible that using Euros as a denomination on the Internet could enhance the use of Internet payment systems within Europe.

9. Conclusions

The technologies for electronic payments are now well understood, and many companies are offering solutions into the market-place, which is currently dominated by card based schemes. The general awareness of the security available on the Internet means that users' fears about keying in their credit card details over the Internet are now subsiding.

Progress in integration between catalogue suppliers and payment suppliers means that it is becoming easier for retailers to set up Web sites to handle on-line payments. Future schemes offering different payment mechanisms are likely to be released into the market-place as new business models demand payments that cannot be made using card-based systems.

References

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Paul Putland joined BT Laboratories in 1985 after graduating in Physics from the University of Exeter. He initially worked on the reliability and analysis of semiconductor devices, and gained an MSc in the physics of laser communications from Essex University in 1992.

In 1992 he joined the applications unit within Applied Research and Technology (ART) to work on a business re-engineering project.

Since 1995, he has been working in the area of electronic commerce and the Internet, initially involved in setting up the eCommerce test bed, and more recently as manager of the BT Array trial.

Chris Ward has over 15 years’ experience of designing and delivering telco and utility billing systems. Over the past 4 years he has been leading on the design and delivery of OSS solutions for BT’s growing Internet, multimedia and eBusiness portfolio.

He has most recently been involved with the selection of an externally sourced rating, provisioning, invoicing and procurement card component for BT’s strategic eBusiness platform.

Alan Jackson currently works in the eCommerce unit, having joined BT in 1975 as an apprentice. After graduating from UMIST in 1983 with a BSc Hons in Electronics he joined BT Laboratories (BTL) to work on analogue transmission circuits and telephone design. He has had a varied technical career involving analogue terminal design covering a wide range of responsibilities from specification and design to production engineering. Most recent work activities have been related to smart cards including phone card schemes, smart card security, Mondex on the Internet, Multos card technology and European standards. He is a member of the IC Card Group of Experts at BTL which proactively addresses smart card security for UK and European schemes managed by BT.

Chris Trollope joined BT in 1985 after graduating in Electronic Engineering from Swansea University and then working as a consultant engineer with WS Atkins. He has worked in the information systems area since joining. Initially providing software development for systems to connect and manage international switches. Then moving through systems strategy and providing systems design and delivery in the MIS and data services. He became involved in Internet and multimedia services in 1995 providing design, delivery, business and consulting services.